

# Dual Stack IP

**Objective:** My objective in this Virtual lab was to demonstrate how basic IPv6/IPv4 configuration is set up. For this topology I made a 2-core layer design with ROAS configuration, holding 3 VLANs per building. The reason for dual IP configuration was to allow easy ping/SSH access to the network devices. I did create 2 loopback interfaces, one for IPv4 and the other for IPv6. As for the DHCP I allowed the end host to take advantage of RA to get the default gateway and their own unique IPv6 address. I also did include a DNS server to show off AAAA records. To make sure the network has full connectivity I configured OSPFv2 and OSPFv3. Lastly, I configured etherchannels to improve the speed of the network.

**Equipment:** (6) 2911 Routers, (13) 2960 Switches, and (25) PC's, and (1) Server

## Key Steps:

- a. Started with the private IPv6 address of FD45:93AC:1::/64 and each VLAN I went up +1 per subnet
- b. Make a set of 3 VLANs per network
- c. Make ROAS configuration per VLAN
- d. Create etherchannels between each set of switches to improve the speed
- e. Assign the first IPv6 available address to each VLAN in order to have static IPs and allow RA to give out the default gateway and network address prefix
- f. For the P2P links between routers I used FD45:93AC:100::/127 and each P2P I went +1 for subnet identifier
- g. Created an IPv6 loopback interface on each router with the prefix of FD45:93AC:200::/128 and I went +1 on the host section to allow easy ping/DNS records
- h. Create a DNS server
- i. Configured OSPFv3 per interface to allow full IPv6 connectivity
- j. Created an IPv4 loopback address on each router device for easy access
- k. For the P2P links between the network devices I used 192.168.1.0/30 to allow full IPv4 connectivity
- l. Each switch falls under the 192.168.2.0/29 network with the VLAN of 200
- m. Created a single subnet for management with the network 192.168.3.0/30 with a VLAN of 300

**Drawbacks :**

The drawbacks from this network design is that there is no way for a 3-core layer design to exist due to Packet Tracer limitations not allowing layer 3 switches to have a IPv6 SVI as the default gateway. Since I wanted to keep the design simple I don't have any failback routes so if one connection fails then a whole department would lose complete communication from everything. I also don't have any security features which makes my network easily breachable.

## Dual Stack IP (interfaces)

|            |         |                |            |
|------------|---------|----------------|------------|
| Switch-A-1 | F0/1    | Port-Channel 1 | Switch-A-3 |
| Switch-A-1 | F0/2    | Port-Channel 1 | Switch-A-3 |
| Switch-A-1 | F0/3    | Port-Channel 1 | Switch-A-3 |
| Switch-A-1 | F0/4    | Port-Channel 2 | Switch-A-2 |
| Switch-A-1 | F0/5    | Port-Channel 2 | Switch-A-2 |
| Switch-A-1 | F0/6    | Port-Channel 2 | Switch-A-2 |
| Switch-A-1 | F0/7    | PC 0           | RS         |
| Switch-A-1 | F0/8    | PC 1           | RS         |
| Switch-A-1 | F0/9    | PC 2           | RS         |
| Switch-A-1 | SVI 300 | 192.168.2.1    | Router-A   |
| Switch-A-2 | F0/1    | Port-Channel 3 | Switch-A-3 |
| Switch-A-2 | F0/2    | Port-Channel 3 | Switch-A-3 |
| Switch-A-2 | F0/3    | Port-Channel 3 | Switch-A-3 |
| Switch-A-2 | F0/4    | Port-Channel 2 | Switch-A-1 |
| Switch-A-2 | F0/5    | Port-Channel 2 | Switch-A-1 |
| Switch-A-2 | F0/6    | Port-Channel 2 | Switch-A-1 |
| Switch-A-2 | F0/7    | PC 3           | RS         |
| Switch-A-2 | F0/8    | PC 4           | RS         |
| Switch-A-2 | F0/9    | PC 5           | RS         |
| Switch-A-2 | SVI 300 | 192.168.2.2    | Router-A   |
| Switch-A-3 | F0/1    | Trunkport      | Router-A   |
| Switch-A-3 | F0/2    | Port-Channel 1 | Switch-A-1 |
| Switch-A-3 | F0/3    | Port-Channel 1 | Switch-A-1 |
| Switch-A-3 | F0/4    | Port-Channel 1 | Switch-A-1 |
| Switch-A-3 | F0/5    | Port-Channel 3 | Switch-A-2 |
| Switch-A-3 | F0/6    | Port-Channel 3 | Switch-A-2 |
| Switch-A-3 | F0/7    | Port-Channel 3 | Switch-A-2 |
| Switch-A-3 | SVI 300 | 192.168.2.3    | Router-A   |
| Switch-B-1 | F0/1    | Port-Channel 1 | Switch-B-3 |
| Switch-B-1 | F0/2    | Port-Channel 1 | Switch-B-3 |
| Switch-B-1 | F0/3    | Port-Channel 1 | Switch-B-3 |
| Switch-B-1 | F0/4    | Port-Channel 2 | Switch-B-2 |
| Switch-B-1 | F0/5    | Port-Channel 2 | Switch-B-2 |
| Switch-B-1 | F0/6    | Port-Channel 2 | Switch-B-2 |
| Switch-B-1 | F0/7    | PC 6           | RS         |
| Switch-B-1 | F0/8    | PC 7           | RS         |

|            |         |                |            |
|------------|---------|----------------|------------|
| Switch-B-1 | F0/9    | PC 8           | RS         |
| Switch-B-1 | SVI 300 | 192.168.2.9    | Router-B   |
| Switch-B-2 | F0/1    | Port-Channel 3 | Switch-B-3 |
| Switch-B-2 | F0/2    | Port-Channel 3 | Switch-B-3 |
| Switch-B-2 | F0/3    | Port-Channel 3 | Switch-B-3 |
| Switch-B-2 | F0/4    | Port-Channel 2 | Switch-B-1 |
| Switch-B-2 | F0/5    | Port-Channel 2 | Switch-B-1 |
| Switch-B-2 | F0/6    | Port-Channel 2 | Switch-B-1 |
| Switch-B-2 | F0/7    | PC 9           | RS         |
| Switch-B-2 | F0/8    | PC 10          | RS         |
| Switch-B-2 | F0/9    | PC 11          | RS         |
| Switch-B-2 | SVI 300 | 192.168.2.10   | Router-B   |
| Switch-B-3 | F0/1    | Trunkport      | Router-B   |
| Switch-B-3 | F0/2    | Port-Channel 1 | Switch-B-1 |
| Switch-B-3 | F0/3    | Port-Channel 1 | Switch-B-1 |
| Switch-B-3 | F0/4    | Port-Channel 1 | Switch-B-1 |
| Switch-B-3 | F0/5    | Port-Channel 3 | Switch-B-2 |
| Switch-B-3 | F0/6    | Port-Channel 3 | Switch-B-2 |
| Switch-B-3 | F0/7    | Port-Channel 3 | Switch-B-2 |
| Switch-B-3 | SVI 300 | 192.168.2.11   | Router-B   |
| Switch-C-1 | F0/1    | Port-Channel 1 | Switch-C-3 |
| Switch-C-1 | F0/2    | Port-Channel 1 | Switch-C-3 |
| Switch-C-1 | F0/3    | Port-Channel 1 | Switch-C-3 |
| Switch-C-1 | F0/4    | Port-Channel 2 | Switch-C-2 |
| Switch-C-1 | F0/5    | Port-Channel 2 | Switch-C-2 |
| Switch-C-1 | F0/6    | Port-Channel 2 | Switch-C-2 |
| Switch-C-1 | F0/7    | PC 12          | RS         |
| Switch-C-1 | F0/8    | PC 13          | RS         |
| Switch-C-1 | F0/9    | PC 14          | RS         |
| Switch-C-1 | SVI 300 | 192.168.2.17   |            |
| Switch-C-2 | F0/1    | Port-Channel 3 | Switch-C-3 |
| Switch-C-2 | F0/2    | Port-Channel 3 | Switch-C-3 |
| Switch-C-2 | F0/3    | Port-Channel 3 | Switch-C-3 |
| Switch-C-2 | F0/4    | Port-Channel 2 | Switch-C-1 |
| Switch-C-2 | F0/5    | Port-Channel 2 | Switch-C-1 |
| Switch-C-2 | F0/6    | Port-Channel 2 | Switch-C-1 |

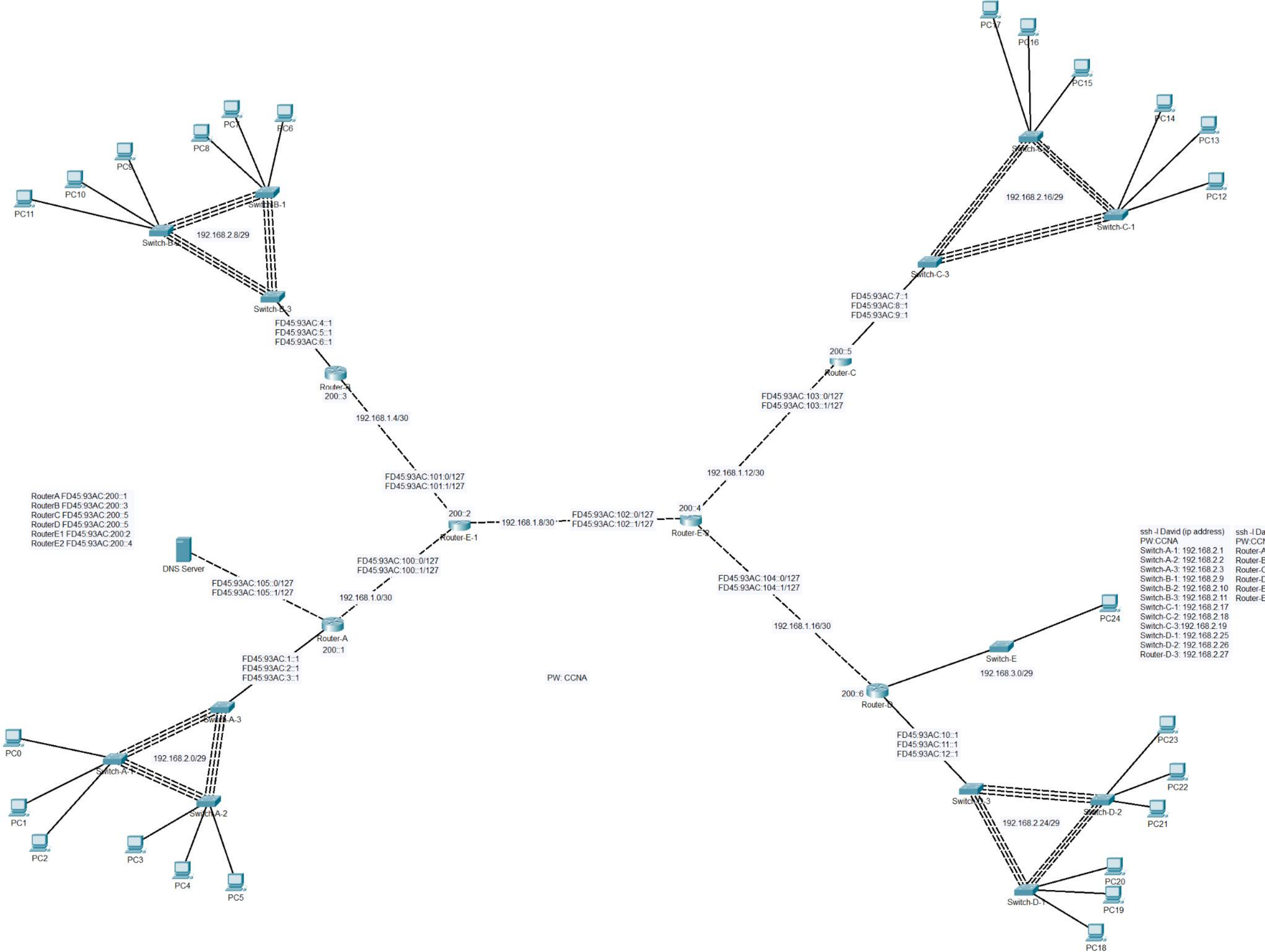
|            |         |                |            |
|------------|---------|----------------|------------|
| Switch-C-2 | F0/7    | PC 15          | RS         |
| Switch-C-2 | F0/8    | PC 16          | RS         |
| Switch-C-2 | F0/9    | PC 17          | RS         |
| Switch-C-2 | SVI 300 | 192.168.2.18   | Router-C   |
| Switch-C-3 | F0/1    | Trunkport      | Router-C   |
| Switch-C-3 | F0/2    | Port-Channel 1 | Switch-C-1 |

|            |         |                |            |
|------------|---------|----------------|------------|
| Switch-C-3 | F0/3    | Port-Channel 1 | Switch-C-1 |
| Switch-C-3 | F0/4    | Port-Channel 1 | Switch-C-1 |
| Switch-C-3 | F0/5    | Port-Channel 3 | Switch-C-2 |
| Switch-C-3 | F0/6    | Port-Channel 3 | Switch-C-2 |
| Switch-C-3 | F0/7    | Port-Channel 3 | Switch-C-2 |
| Switch-C-3 | SVI 300 | 192.168.2.19   | Router-C   |
| Switch-D-1 | F0/1    | Port-Channel 1 | Switch-D-3 |
| Switch-D-1 | F0/2    | Port-Channel 1 | Switch-D-3 |
| Switch-D-1 | F0/3    | Port-Channel 1 | Switch-D-3 |
| Switch-D-1 | F0/4    | Port-Channel 2 | Switch-D-2 |
| Switch-D-1 | F0/5    | Port-Channel 2 | Switch-D-2 |
| Switch-D-1 | F0/6    | Port-Channel 2 | Switch-D-2 |
| Switch-D-1 | F0/7    | PC 18          | RS         |
| Switch-D-1 | F0/8    | PC 19          | RS         |
| Switch-D-1 | F0/9    | PC 20          | RS         |
| Switch-D-1 | SVI 300 | 192.168.2.25   | Router-D   |
| Switch-D-2 | F0/1    | Port-Channel 3 | Switch-D-3 |
| Switch-D-2 | F0/2    | Port-Channel 3 | Switch-D-3 |
| Switch-D-2 | F0/3    | Port-Channel 3 | Switch-D-3 |
| Switch-D-2 | F0/4    | Port-Channel 2 | Switch-D-1 |
| Switch-D-2 | F0/5    | Port-Channel 2 | Switch-D-1 |
| Switch-D-2 | F0/6    | Port-Channel 2 | Switch-D-1 |
| Switch-D-2 | F0/7    | PC 21          | RS         |
| Switch-D-2 | F0/8    | PC 22          | RS         |
| Switch-D-2 | F0/9    | PC 23          | RS         |
| Switch-D-2 | SVI 300 | 192.168.2.26   | Router-D   |
| Switch-D-3 | F0/1    | Trunkport      | Router-D   |
| Switch-D-3 | F0/2    | Port-Channel 1 | Switch-D-1 |
| Switch-D-3 | F0/3    | Port-Channel 1 | Switch-D-1 |
| Switch-D-3 | F0/4    | Port-Channel 1 | Switch-D-1 |
| Switch-D-3 | F0/5    | Port-Channel 3 | Switch-D-2 |
| Switch-D-3 | F0/6    | Port-Channel 3 | Switch-D-2 |
| Switch-D-3 | F0/7    | Port-Channel 3 | Switch-D-2 |
| Switch-D-3 | SVI 300 | 192.168.2.27   | Router-D   |

|          |            |                      |                    |
|----------|------------|----------------------|--------------------|
| Switch-E | F0/1       | Default Gateway      | Router-D           |
| Switch-E | F0/2       | 192.168.3.1          | PC 24              |
| Router-A | G0/0.10    | ROAS                 | FD45:93AC:1::1/64  |
| Router-A | G0/0.20    | ROAS                 | FD45:93AC:2::1/64  |
| Router-A | G0/0.30    | ROAS                 | FD45:93AC:3::1/64  |
| Router-A | G0/0.300   | ROAS                 | 192.168.2.6        |
| Router-A | G0/1       | 192.168.1.1          | Router-E-1         |
| Router-A | G0/1       | FD45:93AC:100::/127  | Router-E-1         |
| Router-A | G0/2       | FD45:93AC:105::/127  | DNS Server         |
| Router-A | Loopback 1 | 1.1.1.1              |                    |
| Router-A | Loopback 2 | FD45:93AC:200::1/128 |                    |
| Router-B | G0/0.40    | ROAS                 | FD45:93AC:4::1/64  |
| Router-B | G0/0.50    | ROAS                 | FD45:93AC:5::1/64  |
| Router-B | G0/0.60    | ROAS                 | FD45:93AC:6::1/64  |
| Router-B | G0/0.300   | ROAS                 | 192.168.2.14       |
| Router-B | G0/1       | FD45:93AC:101::/127  | Router-E-1         |
| Router-B | G0/1       | 192.168.1.5          | Router-E-1         |
| Router-B | Loopback 1 | 3.3.3.3              |                    |
| Router-B | Loopback 2 | FD45:93AC:200::3/128 |                    |
| Router-C | G0/0.70    | ROAS                 | FD45:93AC:7::1/64  |
| Router-C | G0/0.80    | ROAS                 | FD45:93AC:8::1/64  |
| Router-C | G0/0.90    | ROAS                 | FD45:93AC:9::1/64  |
| Router-C | G0/0.300   | ROAS                 | 192.168.2.22       |
| Router-C | G0/1       | 192.168.1.13         | Router-E-2         |
| Router-C | G0/1       | FD45:93AC:103::1/127 | Router-E-2         |
| Router-C | Loopback 1 | 5.5.5.5              |                    |
| Router-C | Loopback 2 | FD45:93AC:200::5/128 |                    |
| Router-D | G0/0.100   | ROAS                 | FD45:93AC:10::1/64 |
| Router-D | G0/0.110   | ROAS                 | FD45:93AC:11::1/64 |
| Router-D | G0/0.120   | ROAS                 | FD45:93AC:12::1/64 |
| Router-D | G0/0.300   | ROAS                 | 192.168.2.30       |

|            |            |                      |             |
|------------|------------|----------------------|-------------|
| Router-D   | G0/1       | 192.168.1.18         | Router-E-2  |
| Router-D   | G0/1       | FD45:93AC:104::1/127 | Router-E-2  |
| Router-D   | G0/2       | Default Gateway      | 192.168.3.6 |
| Router-D   | Loopback 1 | 6.6.6.6              |             |
| Router-D   | Loopback 2 | FD45:93AC:200::6/128 |             |
| Router-E-1 | G0/0       | 192.168.1.2          | Router-E-2  |
| Router-E-1 | G0/0       | FD45:93AC:100::1/127 | Router-E-2  |
| Router-E-1 | G0/1       | 192.168.1.6          | Router-E-2  |
| Router-E-1 | G0/1       | FD45:93AC:101::1/127 | Router-E-2  |
| Router-E-1 | G0/2       | 192.168.1.9          | Router-E-2  |
| Router-E-1 | G0/2       | FD45:93AC:102::1/127 | Router-E-2  |
| Router-E-1 | Loopback 1 | 2.2.2.2              |             |
| Router-E-1 | Loopback 2 | FD45:93AC:200::2/128 |             |
| Router-E-2 | G0/0       | 192.168.1.17         | Router-E-1  |
| Router-E-2 | G0/0       | FD45:93AC:104::1/127 | Router-E-1  |
| Router-E-2 | G0/1       | 192.168.1.14         | Router-E-1  |
| Router-E-2 | G0/1       | FD45:93AC:103::1/127 | Router-E-1  |
| Router-E-2 | G0/2       | 192.168.1.10         | Router-E-1  |
| Router-E-2 | G0/2       | FD45:93AC:102::1/127 | Router-E-1  |
| Router-E-2 | Loopback 1 | 4.4.4.4              |             |
| Router-E-2 | Loopback 2 | FD45:93AC:200::4/128 |             |
| DNS Server | F0         | FD45:93AC:105::1/127 | Router-A    |







# Switch-A-1:

```
!  
hostname Switch-A-1  
!  
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
ip domain-name DavidsLab  
!  
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
!  
interface Port-channel1  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 10,20,30,300  
  switchport mode trunk  
!  
interface Port-channel2  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 10,20,30,300  
  switchport mode trunk  
!  
interface FastEthernet0/1  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 10,20,30,300  
  switchport mode trunk  
  channel-group 1 mode desirable  
!  
interface FastEthernet0/2  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 10,20,30,300  
  switchport mode trunk  
  channel-group 1 mode desirable  
!  
interface FastEthernet0/3  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 10,20,30,300  
  switchport mode trunk  
  channel-group 1 mode desirable  
!  
interface FastEthernet0/4  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 10,20,30,300  
  switchport mode trunk  
  channel-group 2 mode desirable  
!  
interface FastEthernet0/5  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 10,20,30,300  
  switchport mode trunk  
  channel-group 2 mode desirable  
!  
interface FastEthernet0/6  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 10,20,30,300  
  switchport mode trunk  
  channel-group 2 mode desirable  
!
```

```

interface FastEthernet0/7
switchport access vlan 10
switchport mode access
!
interface FastEthernet0/8
switchport access vlan 20
switchport mode access
!
interface FastEthernet0/9
switchport access vlan 30
switchport mode access
!
interface FastEthernet0/10
switchport access vlan 10
switchport mode access
!
interface FastEthernet0/11
switchport access vlan 20
switchport mode access
!
interface FastEthernet0/12
switchport access vlan 30
switchport mode access
!
interface Vlan1
no ip address
shutdown
!
interface Vlan300
ip address 192.168.2.1 255.255.255.248
!
ip default-gateway 192.168.2.6
!
access-list 1 permit host 192.168.3.1
line con 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
end

```

## Switch-A-2:

```

!
hostname Switch-A-2
!
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip domain-name DavidsLab
!
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
spanning-tree mode pvst

```

```
spanning-tree extend system-id
!
interface Port-channel2
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,300
switchport mode trunk
!
interface Port-channel3
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,300
switchport mode trunk
!
interface FastEthernet0/1
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,300
switchport mode trunk
channel-group 3 mode desirable
!
interface FastEthernet0/2
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,300
switchport mode trunk
channel-group 3 mode desirable
!
interface FastEthernet0/3
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,300
switchport mode trunk
channel-group 3 mode desirable
!
interface FastEthernet0/4
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,300
switchport mode trunk
channel-group 2 mode desirable
!
interface FastEthernet0/5
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,300
switchport mode trunk
channel-group 2 mode desirable
!
interface FastEthernet0/6
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,300
switchport mode trunk
channel-group 2 mode desirable
!
interface FastEthernet0/7
switchport access vlan 10
switchport mode access
!
interface FastEthernet0/8
switchport access vlan 20
switchport mode access
!
interface FastEthernet0/9
switchport access vlan 30
switchport mode access
!
interface FastEthernet0/10
```

```

switchport access vlan 10
switchport mode access
!
interface FastEthernet0/11
switchport access vlan 20
switchport mode access
!
interface FastEthernet0/12
switchport access vlan 30
switchport mode access
!
interface Vlan1
no ip address
shutdown
!
interface Vlan300
ip address 192.168.2.2 255.255.255.248
!
ip default-gateway 192.168.2.6
!
access-list 1 permit host 192.168.3.1
line con 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
end

```

## Switch-A-3:

```

!
hostname Switch-A-3
!
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip domain-name DavidsLab
!
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
spanning-tree mode pvst
spanning-tree extend system-id
spanning-tree vlan 10,20,30,300 priority 24576
!
interface Port-channel1
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,300
switchport mode trunk
!
interface Port-channel3
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,300
switchport mode trunk
!

```

```
interface FastEthernet0/1
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,300
switchport mode trunk
!
interface FastEthernet0/2
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,300
switchport mode trunk
channel-group 1 mode desirable
!
interface FastEthernet0/3
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,300
switchport mode trunk
channel-group 1 mode desirable
!
interface FastEthernet0/4
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,300
switchport mode trunk
channel-group 1 mode desirable
!
interface FastEthernet0/5
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,300
switchport mode trunk
channel-group 3 mode desirable
!
interface FastEthernet0/6
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,300
switchport mode trunk
channel-group 3 mode desirable
!
interface FastEthernet0/7
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,300
switchport mode trunk
channel-group 3 mode desirable
!
interface Vlan1
no ip address
shutdown
!
interface Vlan300
ip address 192.168.2.3 255.255.255.248
!
ip default-gateway 192.168.2.6
!
access-list 1 permit host 192.168.3.1
line con 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
```

```
!  
end
```

## Switch-B-1:

```
!  
hostname Switch-B-1  
!  
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
ip domain-name DavidsLab  
!  
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
spanning-tree vlan 40 priority 20480  
!  
interface Port-channel1  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,300  
switchport mode trunk  
!  
interface Port-channel2  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,300  
switchport mode trunk  
!  
interface FastEthernet0/1  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,300  
switchport mode trunk  
channel-group 1 mode desirable  
!  
interface FastEthernet0/2  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,300  
switchport mode trunk  
channel-group 1 mode desirable  
!  
interface FastEthernet0/3  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,300  
switchport mode trunk  
channel-group 1 mode desirable  
!  
interface FastEthernet0/4  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,300  
switchport mode trunk  
channel-group 2 mode desirable  
!  
interface FastEthernet0/5  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,300  
switchport mode trunk  
channel-group 2 mode desirable
```



```

!
interface FastEthernet0/6
 switchport trunk native vlan 330
 switchport trunk allowed vlan 40,50,60,300
 switchport mode trunk
 channel-group 2 mode desirable
!
interface FastEthernet0/7
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/8
 switchport access vlan 50
 switchport mode access
!
interface FastEthernet0/9
 switchport access vlan 60
 switchport mode access
!
interface FastEthernet0/10
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/11
 switchport access vlan 50
 switchport mode access
!
interface FastEthernet0/12
 switchport access vlan 60
 switchport mode access
!
interface Vlan1
 no ip address
 shutdown
!
interface Vlan300
 ip address 192.168.2.9 255.255.255.248
!
ip default-gateway 192.168.2.14
!
access-list 1 permit host 192.168.3.1
line con 0
!
line vty 0 4
 access-class 1 in
 login local
 transport input ssh
line vty 5 15
 access-class 1 in
 login local
 transport input ssh
!
end

```

## Switch-B-2:

```

!
hostname Switch-B-2
!

```

```
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip domain-name DavidsLab
!
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
spanning-tree mode pvst
spanning-tree extend system-id
spanning-tree vlan 40 priority 16384
!
interface Port-channel2
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60,300
switchport mode trunk
!
interface Port-channel3
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60,300
switchport mode trunk
!
interface FastEthernet0/1
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60,300
switchport mode trunk
channel-group 3 mode desirable
!
interface FastEthernet0/2
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60,300
switchport mode trunk
channel-group 3 mode desirable
!
interface FastEthernet0/3
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60,300
switchport mode trunk
channel-group 3 mode desirable
!
interface FastEthernet0/4
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60,300
switchport mode trunk
channel-group 2 mode desirable
!
interface FastEthernet0/5
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60,300
switchport mode trunk
channel-group 2 mode desirable
!
interface FastEthernet0/6
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60,300
switchport mode trunk
channel-group 2 mode desirable
!
interface FastEthernet0/7
switchport access vlan 40
switchport mode access
!
interface FastEthernet0/8
```

```

switchport access vlan 50
switchport mode access
!
interface FastEthernet0/9
switchport access vlan 60
switchport mode access
!
interface FastEthernet0/10
switchport access vlan 40
switchport mode access
!
interface FastEthernet0/11
switchport access vlan 50
switchport mode access
!
interface FastEthernet0/12
switchport access vlan 60
switchport mode access
!
interface Vlan1
no ip address
shutdown
!
interface Vlan300
ip address 192.168.2.10 255.255.255.248
!
ip default-gateway 192.168.2.14
!
access-list 1 permit host 192.168.3.1
line con 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
end

```

## Switch-B-3:

```

!
hostname Switch-B-3
!
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip domain-name DavidsLab
!
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
spanning-tree mode pvst
spanning-tree extend system-id
spanning-tree vlan 40 priority 12288
spanning-tree vlan 50,60,300 priority 24576
!
interface Port-channel1

```

```
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60,300
switchport mode trunk
!
interface Port-channel3
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60,300
switchport mode trunk
!
interface FastEthernet0/1
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60,300
switchport mode trunk
!
interface FastEthernet0/2
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60,300
switchport mode trunk
channel-group 1 mode desirable
!
interface FastEthernet0/3
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60,300
switchport mode trunk
channel-group 1 mode desirable
!
interface FastEthernet0/4
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60,300
switchport mode trunk
channel-group 1 mode desirable
!
interface FastEthernet0/5
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60,300
switchport mode trunk
channel-group 3 mode desirable
!
interface FastEthernet0/6
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60,300
switchport mode trunk
channel-group 3 mode desirable
!
interface FastEthernet0/7
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60,300
switchport mode trunk
channel-group 3 mode desirable
!
interface Vlan1
no ip address
shutdown
!
interface Vlan300
ip address 192.168.2.11 255.255.255.248
!
ip default-gateway 192.168.2.14
!
access-list 1 permit host 192.168.3.1
line con 0
```

```
!  
line vty 0 4  
access-class 1 in  
login local  
transport input ssh  
line vty 5 15  
access-class 1 in  
login local  
transport input ssh  
!  
end
```

## Switch-C-1:

```
!  
hostname Switch-C-1  
!  
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
ip domain-name DavidsLab  
!  
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
!  
interface Port-channel1  
switchport trunk native vlan 330  
switchport trunk allowed vlan 70,80,90,300  
switchport mode trunk  
!  
interface Port-channel2  
switchport trunk native vlan 330  
switchport trunk allowed vlan 70,80,90,300  
switchport mode trunk  
!  
interface FastEthernet0/1  
switchport trunk native vlan 330  
switchport trunk allowed vlan 70,80,90,300  
switchport mode trunk  
channel-group 1 mode desirable  
!  
interface FastEthernet0/2  
switchport trunk native vlan 330  
switchport trunk allowed vlan 70,80,90,300  
switchport mode trunk  
channel-group 1 mode desirable  
!  
interface FastEthernet0/3  
switchport trunk native vlan 330  
switchport trunk allowed vlan 70,80,90,300  
switchport mode trunk  
channel-group 1 mode desirable  
!  
interface FastEthernet0/4  
switchport trunk native vlan 330  
switchport trunk allowed vlan 70,80,90,300  
switchport mode trunk  
channel-group 2 mode desirable  
!
```

```
interface FastEthernet0/5
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,300
switchport mode trunk
channel-group 2 mode desirable
!
interface FastEthernet0/6
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,300
switchport mode trunk
channel-group 2 mode desirable
!
interface FastEthernet0/7
switchport access vlan 70
switchport mode access
!
interface FastEthernet0/8
switchport access vlan 80
switchport mode access
!
interface FastEthernet0/9
switchport access vlan 90
switchport mode access
!
interface FastEthernet0/10
switchport access vlan 70
switchport mode access
!
interface FastEthernet0/11
switchport access vlan 80
switchport mode access
!
interface FastEthernet0/12
switchport access vlan 90
switchport mode access
!
interface Vlan1
no ip address
shutdown
!
interface Vlan300
ip address 192.168.2.17 255.255.255.248
!
ip default-gateway 192.168.2.22
!
access-list 1 permit host 192.168.3.1
line con 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
end
```

## Switch-C-2:

```
!  
hostname Switch-C-2  
!  
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
ip domain-name DavidsLab  
!  
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
!  
interface Port-channel2  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 70,80,90,300  
  switchport mode trunk  
!  
interface Port-channel3  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 70,80,90,300  
  switchport mode trunk  
!  
interface FastEthernet0/1  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 70,80,90,300  
  switchport mode trunk  
  channel-group 3 mode desirable  
!  
interface FastEthernet0/2  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 70,80,90,300  
  switchport mode trunk  
  channel-group 3 mode desirable  
!  
interface FastEthernet0/3  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 70,80,90,300  
  switchport mode trunk  
  channel-group 3 mode desirable  
!  
interface FastEthernet0/4  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 70,80,90,300  
  switchport mode trunk  
  channel-group 2 mode desirable  
!  
interface FastEthernet0/5  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 70,80,90,300  
  switchport mode trunk  
  channel-group 2 mode desirable  
!  
interface FastEthernet0/6  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 70,80,90,300  
  switchport mode trunk  
  channel-group 2 mode desirable  
!
```

```

interface FastEthernet0/7
switchport access vlan 70
switchport mode access
!
interface FastEthernet0/8
switchport access vlan 80
switchport mode access
!
interface FastEthernet0/9
switchport access vlan 90
switchport mode access
!
interface FastEthernet0/10
switchport access vlan 70
switchport mode access
!
interface FastEthernet0/11
switchport access vlan 80
switchport mode access
!
interface FastEthernet0/12
switchport access vlan 90
switchport mode access
!
interface Vlan1
no ip address
shutdown
!
interface Vlan300
ip address 192.168.2.18 255.255.255.248
!
ip default-gateway 192.168.2.22
!
access-list 1 permit host 192.168.3.1
line con 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
end

```

## Switch-C-3:

```

!
hostname Switch-C-3
!
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip domain-name DavidsLab
!
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
spanning-tree mode pvst

```



```
spanning-tree extend system-id
spanning-tree vlan 70,80,90,300 priority 24576
!
interface Port-channel1
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,300
switchport mode trunk
!
interface Port-channel3
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,300
switchport mode trunk
!
interface FastEthernet0/1
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,300
switchport mode trunk
!
interface FastEthernet0/2
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,300
switchport mode trunk
channel-group 1 mode desirable
!
interface FastEthernet0/3
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,300
switchport mode trunk
channel-group 1 mode desirable
!
interface FastEthernet0/4
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,300
switchport mode trunk
channel-group 1 mode desirable
!
interface FastEthernet0/5
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,300
switchport mode trunk
channel-group 3 mode desirable
!
interface FastEthernet0/6
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,300
switchport mode trunk
channel-group 3 mode desirable
!
interface FastEthernet0/7
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,300
switchport mode trunk
channel-group 3 mode desirable
!
interface Vlan1
no ip address
shutdown
!
interface Vlan300
ip address 192.168.2.19 255.255.255.248
!
```

```
ip default-gateway 192.168.2.22
!
access-list 1 permit host 192.168.3.1
line con 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
end
```

## Switch-D-1:

```
!
hostname Switch-D-1
!
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip domain-name DavidsLab
!
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface Port-channel1
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,300
switchport mode trunk
!
interface Port-channel2
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,300
switchport mode trunk
!
interface FastEthernet0/1
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,300
switchport mode trunk
channel-group 1 mode desirable
!
interface FastEthernet0/2
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,300
switchport mode trunk
channel-group 1 mode desirable
!
interface FastEthernet0/3
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,300
switchport mode trunk
channel-group 1 mode desirable
```

```
!  
interface FastEthernet0/4  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 100,110,120,300  
  switchport mode trunk  
  channel-group 2 mode desirable  
!  
interface FastEthernet0/5  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 100,110,120,300  
  switchport mode trunk  
  channel-group 2 mode desirable  
!  
interface FastEthernet0/6  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 100,110,120,300  
  switchport mode trunk  
  channel-group 2 mode desirable  
!  
interface FastEthernet0/7  
  switchport access vlan 100  
  switchport mode access  
!  
interface FastEthernet0/8  
  switchport access vlan 110  
  switchport mode access  
!  
interface FastEthernet0/9  
  switchport access vlan 120  
  switchport mode access  
!  
interface FastEthernet0/10  
  switchport access vlan 100  
  switchport mode access  
!  
interface FastEthernet0/11  
  switchport access vlan 110  
  switchport mode access  
!  
interface FastEthernet0/12  
  switchport access vlan 120  
  switchport mode access  
!  
interface Vlan1  
  no ip address  
  shutdown  
!  
interface Vlan300  
  ip address 192.168.2.25 255.255.255.248  
!  
ip default-gateway 192.168.2.30  
!  
access-list 1 permit host 192.168.3.1  
line con 0  
!  
line vty 0 4  
  access-class 1 in  
  login local  
  transport input ssh  
line vty 5 15  
  access-class 1 in
```

```
login local
transport input ssh
!
end
```

## Switch-D-2:

```
!
hostname Switch-D-2
!
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip domain-name DavidsLab
!
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface Port-channel2
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,300
switchport mode trunk
!
interface Port-channel3
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,300
switchport mode trunk
!
interface FastEthernet0/1
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,300
switchport mode trunk
channel-group 3 mode desirable
!
interface FastEthernet0/2
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,300
switchport mode trunk
channel-group 3 mode desirable
!
interface FastEthernet0/3
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,300
switchport mode trunk
channel-group 3 mode desirable
!
interface FastEthernet0/4
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,300
switchport mode trunk
channel-group 2 mode desirable
!
interface FastEthernet0/5
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,300
switchport mode trunk
channel-group 2 mode desirable
!
```

```

interface FastEthernet0/6
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,300
switchport mode trunk
channel-group 2 mode desirable
!
interface FastEthernet0/7
switchport access vlan 100
switchport mode access
!
interface FastEthernet0/8
switchport access vlan 110
switchport mode access
!
interface FastEthernet0/9
switchport access vlan 120
switchport mode access
!
interface FastEthernet0/10
switchport access vlan 100
switchport mode access
!
interface FastEthernet0/11
switchport access vlan 110
switchport mode access
!
interface FastEthernet0/12
switchport access vlan 120
switchport mode access
!
interface Vlan1
no ip address
shutdown
!
interface Vlan300
ip address 192.168.2.26 255.255.255.248
!
ip default-gateway 192.168.2.30
!
access-list 1 permit host 192.168.3.1
line con 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
end

```

## Switch-D-3:

```

!
hostname Switch-D-3
!
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/

```

```
!  
ip domain-name DavidsLab  
!  
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
spanning-tree vlan 100,110,120 priority 24576  
!  
interface Port-channel1  
switchport trunk native vlan 330  
switchport trunk allowed vlan 100,110,120,300  
switchport mode trunk  
!  
interface Port-channel3  
switchport trunk native vlan 330  
switchport trunk allowed vlan 100,110,120,300  
switchport mode trunk  
!  
interface FastEthernet0/1  
switchport trunk native vlan 330  
switchport trunk allowed vlan 100,110,120,300  
switchport mode trunk  
!  
interface FastEthernet0/2  
switchport trunk native vlan 330  
switchport trunk allowed vlan 100,110,120,300  
switchport mode trunk  
channel-group 1 mode desirable  
!  
interface FastEthernet0/3  
switchport trunk native vlan 330  
switchport trunk allowed vlan 100,110,120,300  
switchport mode trunk  
channel-group 1 mode desirable  
!  
interface FastEthernet0/4  
switchport trunk native vlan 330  
switchport trunk allowed vlan 100,110,120,300  
switchport mode trunk  
channel-group 1 mode desirable  
!  
interface FastEthernet0/5  
switchport trunk native vlan 330  
switchport trunk allowed vlan 100,110,120,300  
switchport mode trunk  
channel-group 3 mode desirable  
!  
interface FastEthernet0/6  
switchport trunk native vlan 330  
switchport trunk allowed vlan 100,110,120,300  
switchport mode trunk  
channel-group 3 mode desirable  
!  
interface FastEthernet0/7  
switchport trunk native vlan 330  
switchport trunk allowed vlan 100,110,120,300  
switchport mode trunk  
channel-group 3 mode desirable  
!  
interface Vlan1
```

```
no ip address
shutdown
!
interface Vlan300
ip address 192.168.2.27 255.255.255.248
!
ip default-gateway 192.168.2.30
!
access-list 1 permit host 192.168.3.1
line con 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
end
```

## Switch-E:

```
!
hostname Switch-E
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
switchport access vlan 200
switchport mode access
!
interface FastEthernet0/2
switchport access vlan 200
switchport mode access
!
interface Vlan1
no ip address
shutdown
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
!
end
```

## Router-A:

```
!
hostname Router-A
!
```

```
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip cef
ipv6 unicast-routing
!
no ipv6 cef
!
ipv6 dhcp pool VLAN10
  dns-server FD45:93AC:105::1
!
ipv6 dhcp pool VLAN20
  dns-server FD45:93AC:105::1
!
ipv6 dhcp pool VLAN30
  dns-server FD45:93AC:105::1
!
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
license udi pid CISCO2911/K9 sn FTX1524AH32-
!
ip domain-name DavidsLab
!
spanning-tree mode pvst
!
interface Loopback1
  ip address 1.1.1.1 255.255.255.255
!
interface Loopback2
  no ip address
  ipv6 address FD45:93AC:200::1/128
  ipv6 enable
  ipv6 ospf 1 area 0
!
interface GigabitEthernet0/0
  no ip address
  duplex auto
  speed auto
!
interface GigabitEthernet0/0.10
  encapsulation dot1Q 10
  no ip address
  ipv6 address FD45:93AC:1::1/64
  ipv6 nd other-config-flag
  ipv6 enable
  ipv6 ospf 1 area 0
  ipv6 dhcp server VLAN10
!
interface GigabitEthernet0/0.20
  encapsulation dot1Q 20
  no ip address
  ipv6 address FD45:93AC:2::1/64
  ipv6 nd other-config-flag
  ipv6 enable
  ipv6 ospf 1 area 0
  ipv6 dhcp server VLAN20
!
interface GigabitEthernet0/0.30
  encapsulation dot1Q 30
  no ip address
  ipv6 address FD45:93AC:3::1/64
  ipv6 nd other-config-flag
```



```
ipv6 enable
ipv6 ospf 1 area 0
ipv6 dhcp server VLAN30
!
interface GigabitEthernet0/0.300
encapsulation dot1Q 300
ip address 192.168.2.6 255.255.255.248
!
interface GigabitEthernet0/1
ip address 192.168.1.1 255.255.255.252
duplex auto
speed auto
ipv6 address FD45:93AC:100::/127
ipv6 enable
ipv6 ospf 1 area 0
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
ipv6 address FD45:93AC:105::/127
ipv6 enable
ipv6 ospf 1 area 0
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
log-adjacency-changes
network 1.1.1.1 0.0.0.0 area 0
network 192.168.1.0 0.0.0.3 area 0
network 192.168.2.0 0.0.0.7 area 0
!
ipv6 router ospf 1
router-id 1.1.1.1
log-adjacency-changes
!
ip classless
!
ip flow-export version 9
!
access-list 1 permit host 192.168.3.1
!
line con 0
!
line aux 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
end
```

# Router-B:

```
!  
hostname Router-B  
!  
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
ip cef  
ipv6 unicast-routing  
!  
no ipv6 cef  
!  
ipv6 dhcp pool VLAN40  
  dns-server FD45:93AC:105::1  
!  
ipv6 dhcp pool VLAN50  
  dns-server FD45:93AC:105::1  
!  
ipv6 dhcp pool VLAN60  
  dns-server FD45:93AC:105::1  
!  
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
license udi pid CISCO2911/K9 sn FTX1524MWC1-  
!  
ip domain-name DavidsLab  
!  
spanning-tree mode pvst  
!  
interface Loopback1  
  ip address 3.3.3.3 255.255.255.255  
!  
interface Loopback2  
  no ip address  
  ipv6 address FD45:93AC:200::3/128  
  ipv6 enable  
  ipv6 ospf 1 area 0  
!  
interface GigabitEthernet0/0  
  no ip address  
  duplex auto  
  speed auto  
!  
interface GigabitEthernet0/0.40  
  encapsulation dot1Q 40  
  no ip address  
  ipv6 address FD45:93AC:4::1/64  
  ipv6 nd other-config-flag  
  ipv6 enable  
  ipv6 ospf 1 area 0  
  ipv6 dhcp server VLAN40  
!  
interface GigabitEthernet0/0.50  
  encapsulation dot1Q 50  
  no ip address  
  ipv6 address FD45:93AC:5::1/64  
  ipv6 nd other-config-flag  
  ipv6 enable  
  ipv6 ospf 1 area 0  
  ipv6 dhcp server VLAN50
```

```
!  
interface GigabitEthernet0/0.60  
encapsulation dot1Q 60  
no ip address  
ipv6 address FD45:93AC:6::1/64  
ipv6 nd other-config-flag  
ipv6 enable  
ipv6 ospf 1 area 0  
ipv6 dhcp server VLAN60  
!  
interface GigabitEthernet0/0.300  
encapsulation dot1Q 300  
ip address 192.168.2.14 255.255.255.248  
!  
interface GigabitEthernet0/1  
ip address 192.168.1.5 255.255.255.252  
duplex auto  
speed auto  
ipv6 address FD45:93AC:101::/127  
ipv6 enable  
ipv6 ospf 1 area 0  
!  
interface GigabitEthernet0/2  
no ip address  
duplex auto  
speed auto  
shutdown  
!  
interface Vlan1  
no ip address  
shutdown  
!  
router ospf 1  
log-adjacency-changes  
network 3.3.3.3 0.0.0.0 area 0  
network 192.168.1.4 0.0.0.3 area 0  
network 192.168.2.8 0.0.0.7 area 0  
!  
ipv6 router ospf 1  
router-id 3.3.3.3  
log-adjacency-changes  
!  
ip classless  
!  
ip flow-export version 9  
!  
access-list 1 permit host 192.168.3.1  
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
access-class 1 in  
login local  
transport input ssh  
line vty 5 15  
access-class 1 in  
login local  
transport input ssh  
!
```

end

## Router-C:

```
!  
hostname Router-C  
!  
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
ip cef  
ipv6 unicast-routing  
!  
no ipv6 cef  
!  
ipv6 dhcp pool VLAN70  
  dns-server FD45:93AC:105::1  
!  
ipv6 dhcp pool VLAN80  
  dns-server FD45:93AC:105::1  
!  
ipv6 dhcp pool VLAN90  
  dns-server FD45:93AC:105::1  
!  
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
license udi pid CISCO2911/K9 sn FTX1524M3OV-  
!  
ip domain-name DavidsLab  
!  
spanning-tree mode pvst  
!  
interface Loopback1  
  ip address 5.5.5.5 255.255.255.255  
!  
interface Loopback2  
  no ip address  
  ipv6 address FD45:93AC:200::5/128  
  ipv6 enable  
  ipv6 ospf 1 area 0  
!  
interface GigabitEthernet0/0  
  no ip address  
  duplex auto  
  speed auto  
!  
interface GigabitEthernet0/0.70  
  encapsulation dot1Q 70  
  no ip address  
  ipv6 address FD45:93AC:7::1/64  
  ipv6 nd other-config-flag  
  ipv6 enable  
  ipv6 ospf 1 area 0  
  ipv6 dhcp server VLAN70  
!  
interface GigabitEthernet0/0.80  
  encapsulation dot1Q 80  
  no ip address  
  ipv6 address FD45:93AC:8::1/64  
  ipv6 nd other-config-flag
```

```
ipv6 enable
ipv6 ospf 1 area 0
ipv6 dhcp server VLAN80
!
interface GigabitEthernet0/0.90
encapsulation dot1Q 90
no ip address
ipv6 address FD45:93AC:9::1/64
ipv6 nd other-config-flag
ipv6 enable
ipv6 ospf 1 area 0
ipv6 dhcp server VLAN90
!
interface GigabitEthernet0/0.300
encapsulation dot1Q 300
ip address 192.168.2.22 255.255.255.248
!
interface GigabitEthernet0/1
ip address 192.168.1.13 255.255.255.252
duplex auto
speed auto
ipv6 address FD45:93AC:103::1/127
ipv6 enable
ipv6 ospf 1 area 0
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
log-adjacency-changes
network 192.168.1.12 0.0.0.3 area 0
network 5.5.5.5 0.0.0.0 area 0
network 192.168.2.16 0.0.0.7 area 0
!
ipv6 router ospf 1
router-id 5.5.5.5
log-adjacency-changes
!
ip classless
!
ip flow-export version 9
!
access-list 1 permit host 192.168.3.1
!
line con 0
!
line aux 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
```

```
login local
transport input ssh
!
end
```

## Router-D:

```
!
hostname Router-D
!
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip cef
ipv6 unicast-routing
!
no ipv6 cef
!
ipv6 dhcp pool VLAN100
  dns-server FD45:93AC:105::1
!
ipv6 dhcp pool VLAN110
  dns-server FD45:93AC:105::1
!
ipv6 dhcp pool VLAN120
  dns-server FD45:93AC:105::1
!
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
license udi pid CISCO2911/K9 sn FTX1524HKNF-
!
ip domain-name DavidsLab
!
spanning-tree mode pvst
!
interface Loopback1
  ip address 6.6.6.6 255.255.255.255
!
interface Loopback2
  no ip address
  ipv6 address FD45:93AC:200::6/128
  ipv6 enable
  ipv6 ospf 1 area 0
!
interface GigabitEthernet0/0
  no ip address
  duplex auto
  speed auto
!
interface GigabitEthernet0/0.100
  encapsulation dot1Q 100
  no ip address
  ipv6 address FD45:93AC:10::1/64
  ipv6 nd other-config-flag
  ipv6 enable
  ipv6 ospf 1 area 0
  ipv6 dhcp server VLAN100
!
interface GigabitEthernet0/0.110
  encapsulation dot1Q 110
  no ip address
```

```
ipv6 address FD45:93AC:11::1/64
ipv6 nd other-config-flag
ipv6 enable
ipv6 ospf 1 area 0
ipv6 dhcp server VLAN110
!
interface GigabitEthernet0/0.120
encapsulation dot1Q 120
no ip address
ipv6 address FD45:93AC:12::1/64
ipv6 nd other-config-flag
ipv6 enable
ipv6 ospf 1 area 0
ipv6 dhcp server VLAN120
!
interface GigabitEthernet0/0.300
encapsulation dot1Q 300
ip address 192.168.2.30 255.255.255.248
!
interface GigabitEthernet0/1
ip address 192.168.1.18 255.255.255.252
duplex auto
speed auto
ipv6 address FD45:93AC:104::1/127
ipv6 enable
ipv6 ospf 1 area 0
!
interface GigabitEthernet0/2
ip address 192.168.3.6 255.255.255.248
duplex auto
speed auto
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
log-adjacency-changes
network 192.168.1.16 0.0.0.3 area 0
network 6.6.6.6 0.0.0.0 area 0
network 192.168.3.0 0.0.0.7 area 0
network 192.168.2.24 0.0.0.7 area 0
!
ipv6 router ospf 1
router-id 6.6.6.6
log-adjacency-changes
!
ip classless
!
ip flow-export version 9
!
access-list 1 permit host 192.168.3.1
!
line con 0
!
line aux 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
```

```
line vty 5 15
access-class 1 in
login local
transport input ssh
!!
end
```

## Router-E-1:

```
!
hostname Router-E-1
!
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip cef
ipv6 unicast-routing
!
no ipv6 cef
!
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
license udi pid CISCO2911/K9 sn FTX1524C1WQ-
!
ip domain-name DavidsLab
!
spanning-tree mode pvst
!
interface Loopback1
ip address 2.2.2.2 255.255.255.255
!
interface Loopback2
no ip address
ipv6 address FD45:93AC:200::2/128
ipv6 enable
ipv6 ospf 1 area 0
!
interface GigabitEthernet0/0
ip address 192.168.1.2 255.255.255.252
duplex auto
speed auto
ipv6 address FD45:93AC:100::1/127
ipv6 enable
ipv6 ospf 1 area 0
!
interface GigabitEthernet0/1
ip address 192.168.1.6 255.255.255.252
duplex auto
speed auto
ipv6 address FD45:93AC:101::1/127
ipv6 enable
ipv6 ospf 1 area 0
!
interface GigabitEthernet0/2
ip address 192.168.1.9 255.255.255.252
duplex auto
speed auto
ipv6 address FD45:93AC:102::/127
ipv6 enable
ipv6 ospf 1 area 0
```



```

!
interface Vlan1
no ip address
shutdown
!
router ospf 1
log-adjacency-changes
network 2.2.2.2 0.0.0.0 area 0
network 192.168.1.0 0.0.0.3 area 0
network 192.168.1.4 0.0.0.3 area 0
network 192.168.1.8 0.0.0.3 area 0
!
ipv6 router ospf 1
router-id 2.2.2.2
log-adjacency-changes
!
ip classless
!
ip flow-export version 9
!
access-list 1 permit host 192.168.3.1
!
line con 0
!
line aux 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
end

```

## Router-E-2:

```

!
hostname Router-E-2
!
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip cef
ipv6 unicast-routing
!
no ipv6 cef
!
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
license udi pid CISCO2911/K9 sn FTX1524IGD0-
!
ip domain-name DavidsLab
!
spanning-tree mode pvst
!
interface Loopback1
ip address 4.4.4.4 255.255.255.255

```

```
!  
interface Loopback2  
no ip address  
ipv6 address FD45:93AC:200::4/128  
ipv6 enable  
ipv6 ospf 1 area 0  
!  
interface GigabitEthernet0/0  
ip address 192.168.1.17 255.255.255.252  
duplex auto  
speed auto  
ipv6 address FD45:93AC:104::/127  
ipv6 enable  
ipv6 ospf 1 area 0  
!  
interface GigabitEthernet0/1  
ip address 192.168.1.14 255.255.255.252  
duplex auto  
speed auto  
ipv6 address FD45:93AC:103::/127  
ipv6 enable  
ipv6 ospf 1 area 0  
!  
interface GigabitEthernet0/2  
ip address 192.168.1.10 255.255.255.252  
duplex auto  
speed auto  
ipv6 address FD45:93AC:102::1/127  
ipv6 enable  
ipv6 ospf 1 area 0  
!  
interface Vlan1  
no ip address  
shutdown  
!  
router ospf 1  
log-adjacency-changes  
network 192.168.1.8 0.0.0.3 area 0  
network 4.4.4.4 0.0.0.0 area 0  
network 192.168.1.12 0.0.0.3 area 0  
network 192.168.1.16 0.0.0.3 area 0  
!  
ipv6 router ospf 1  
router-id 4.4.4.4  
log-adjacency-changes  
!  
ip classless  
!  
ip flow-export version 9  
!  
access-list 1 permit host 192.168.3.1  
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
access-class 1 in  
login local  
transport input ssh  
line vty 5 15
```

```
access-class 1 in
login local
transport input ssh
!
end
```